



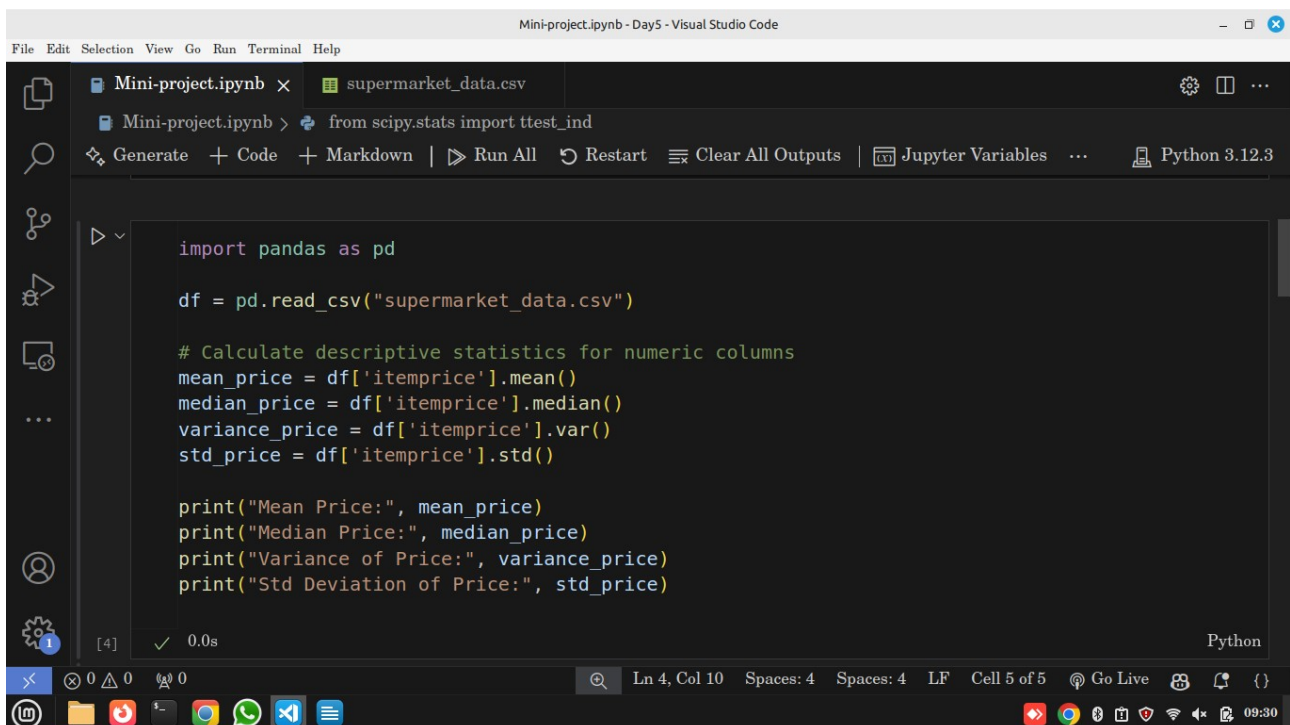
PRACTICALS USING SUPERMARKET DATA

1 DESCRIPTIVE STATISTICS

Concept

Descriptive statistics summarize data using measures like **mean**, **median**, **variance**, and **standard deviation**.

Code



```
Mini-project.ipynb - Day5 - Visual Studio Code
File Edit Selection View Go Run Terminal Help

Mini-project.ipynb x supermarket_data.csv
Mini-project.ipynb > from scipy.stats import ttest_ind
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import pandas as pd

df = pd.read_csv("supermarket_data.csv")

# Calculate descriptive statistics for numeric columns
mean_price = df['itemprice'].mean()
median_price = df['itemprice'].median()
variance_price = df['itemprice'].var()
std_price = df['itemprice'].std()

print("Mean Price:", mean_price)
print("Median Price:", median_price)
print("Variance of Price:", variance_price)
print("Std Deviation of Price:", std_price)
```

Code Explanation

- `.mean()` → Average price
- `.median()` → Middle price
- `.var()` → Measures spread of prices
- `.std()` → How much prices vary on average

Business Insight

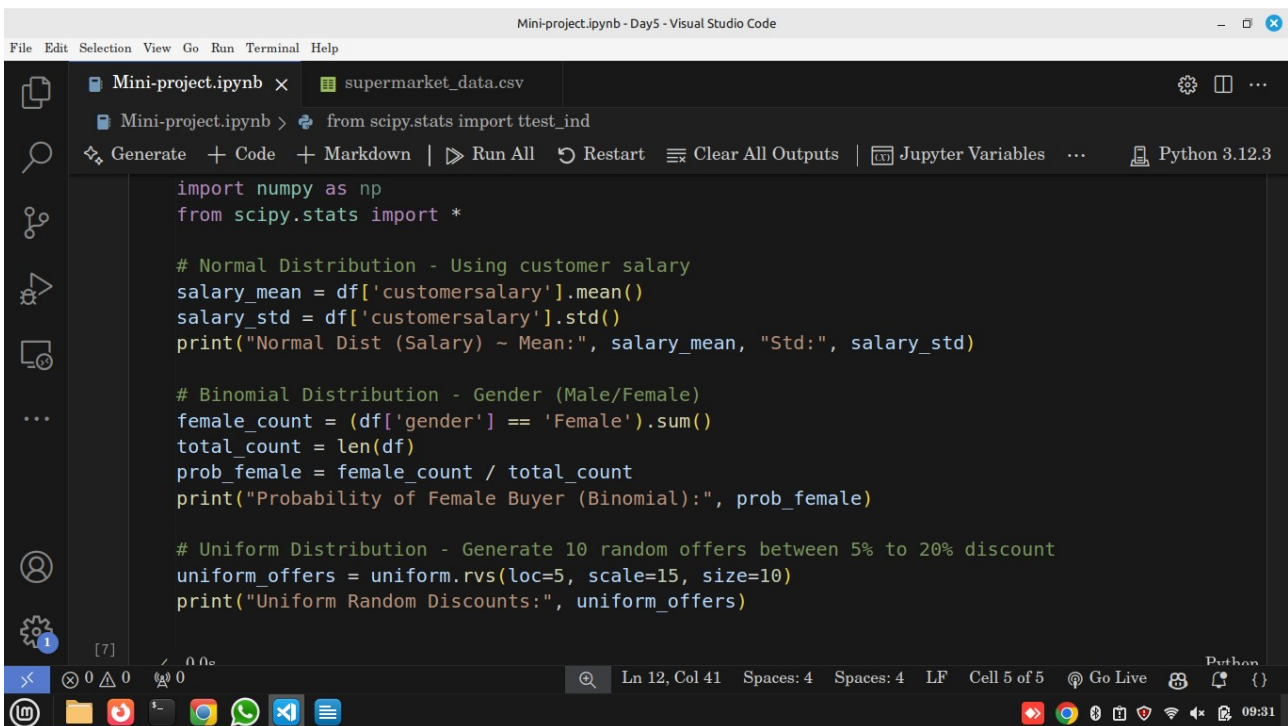
A high **standard deviation** means customers buy products with very different prices. A low value means prices are consistent.

2 PROBABILITY DISTRIBUTIONS (Normal, Binomial, Uniform)

Concept

- **Normal:** Bell-shaped (e.g., customer spending)
- **Binomial:** Success/Failure (e.g., male vs female buyers)
- **Uniform:** Equal chance (e.g., random sampling)

Code



```
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import numpy as np
from scipy.stats import *

# Normal Distribution - Using customer salary
salary_mean = df['customersalary'].mean()
salary_std = df['customersalary'].std()
print("Normal Dist (Salary) ~ Mean:", salary_mean, "Std:", salary_std)

# Binomial Distribution - Gender (Male/Female)
female_count = (df['gender'] == 'Female').sum()
total_count = len(df)
prob_female = female_count / total_count
print("Probability of Female Buyer (Binomial):", prob_female)

# Uniform Distribution - Generate 10 random offers between 5% to 20% discount
uniform_offers = uniform.rvs(loc=5, scale=15, size=10)
print("Uniform Random Discounts:", uniform_offers)

[7]
```

Insight

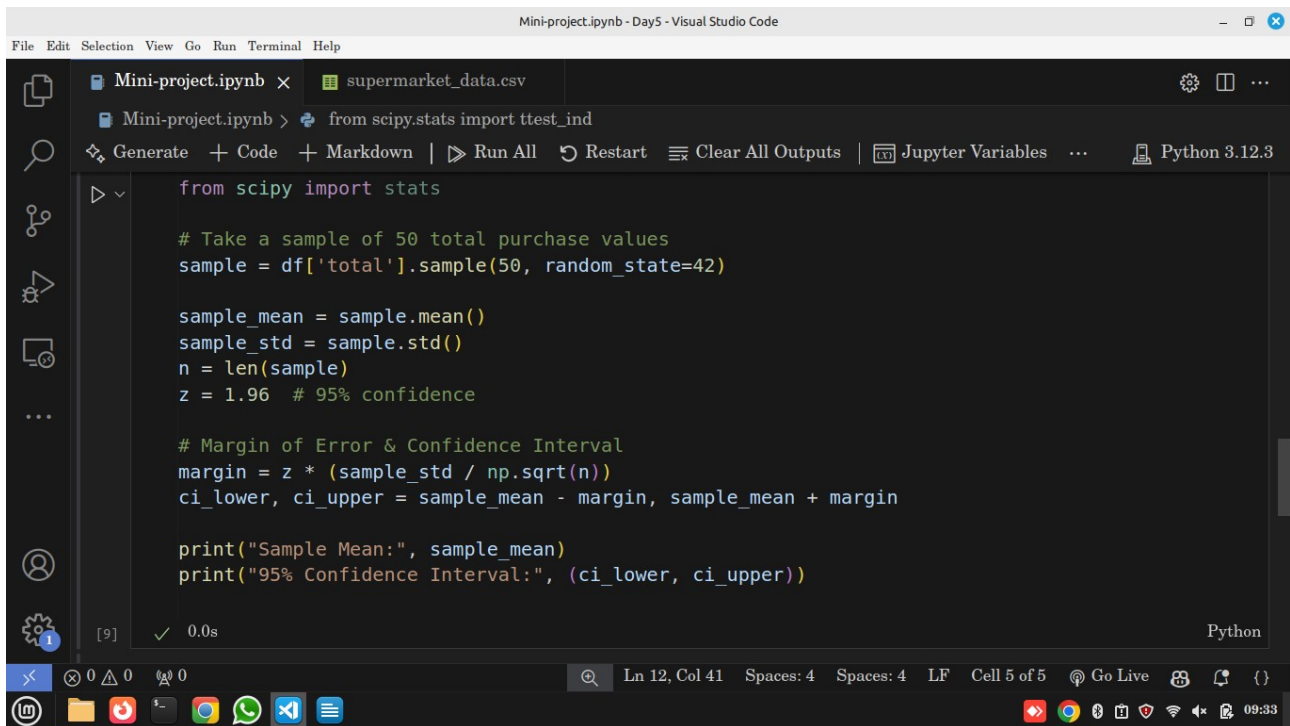
- **Normal:** Salaries cluster around average.
 - **Binomial:** Helps estimate customer demographics.
 - **Uniform:** Used for random promo generation.
-

3 INFERENCEAL STATS (Sampling & Confidence Intervals)

Concept

We use **sample data** to estimate **population parameters** with confidence.

Code



```
Mini-project.ipynb - Day5 - Visual Studio Code
File Edit Selection View Go Run Terminal Help

Mini-project.ipynb x supermarket_data.csv
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from scipy import stats

# Take a sample of 50 total purchase values
sample = df['total'].sample(50, random_state=42)

sample_mean = sample.mean()
sample_std = sample.std()
n = len(sample)
z = 1.96 # 95% confidence

# Margin of Error & Confidence Interval
margin = z * (sample_std / np.sqrt(n))
ci_lower, ci_upper = sample_mean - margin, sample_mean + margin

print("Sample Mean:", sample_mean)
print("95% Confidence Interval:", (ci_lower, ci_upper))

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```

Explanation

- **Sample** → Partial data
- **CI** → Estimated range of true average spending

Insight

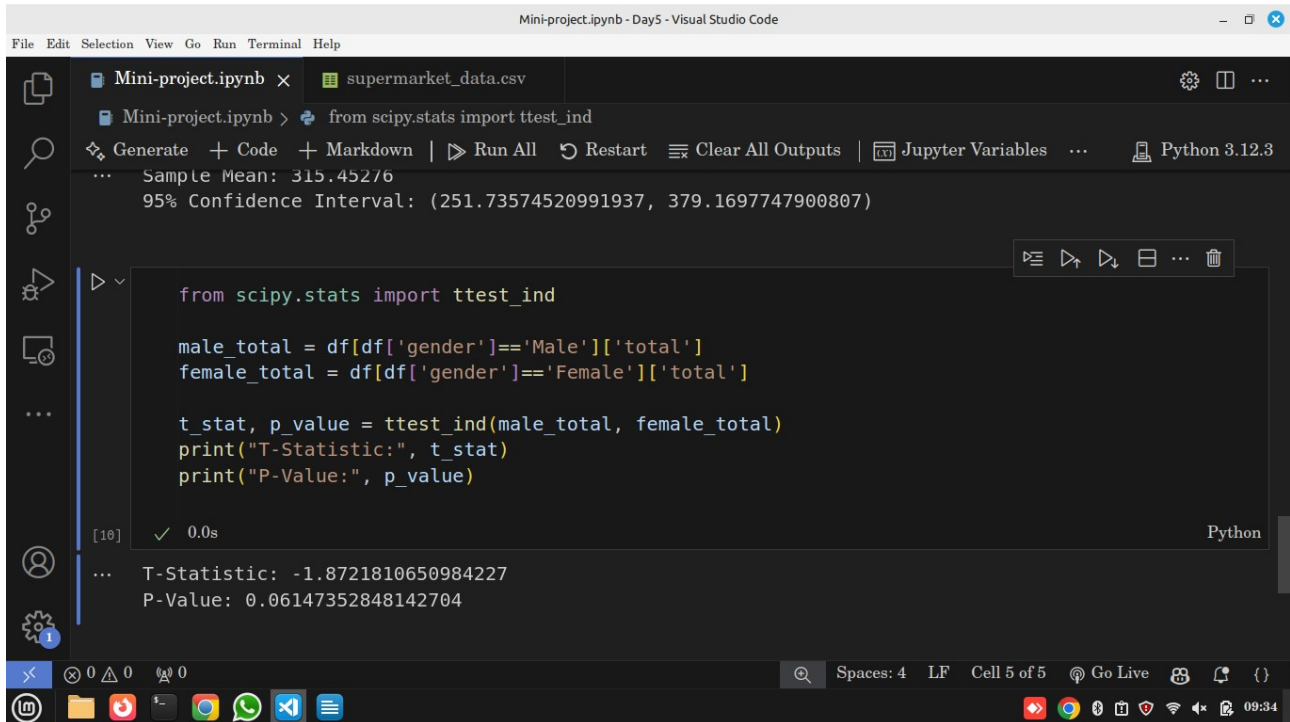
“We are 95% confident that true customer spending lies between these two values.”

4 HYPOTHESIS TESTING (t-test & Chi-Square)

Concept

Used to **check claims** or compare groups.

◆ t-Test Example (Male vs Female Spending)



The screenshot shows a Jupyter Notebook titled 'Mini-project.ipynb' in Visual Studio Code. The notebook is open to a cell containing Python code for a t-test. The code imports 'ttest_ind' from 'scipy.stats', calculates the total spending for males and females from a dataset 'df', and then performs a t-test. The output shows a T-Statistic of -1.8721810650984227 and a P-Value of 0.06147352848142704. The notebook interface includes a file explorer on the left, a terminal at the bottom, and a status bar at the very bottom.

```
from scipy.stats import ttest_ind

male_total = df[df['gender']=='Male']['total']
female_total = df[df['gender']=='Female']['total']

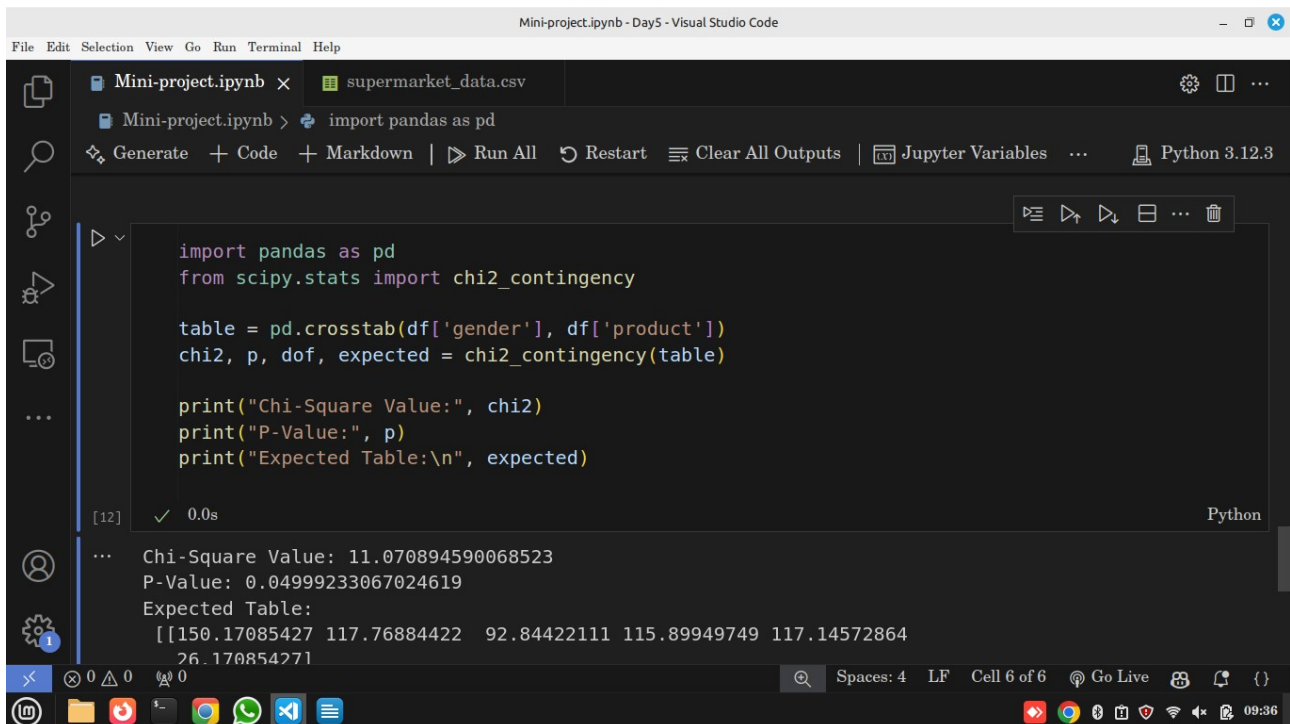
t_stat, p_value = ttest_ind(male_total, female_total)
print("T-Statistic:", t_stat)
print("P-Value:", p_value)
```

[10] ✓ 0.0s Python

... T-Statistic: -1.8721810650984227
P-Value: 0.06147352848142704

Insight: If $p\text{-value} < 0.05$, spending is different between genders.

◆ Chi-Square Test (Gender vs Product Preference)



The screenshot shows a Jupyter Notebook titled 'Mini-project.ipynb' in Visual Studio Code. The notebook is open to a cell containing Python code for a Chi-Square test. The code imports 'pd' from 'pandas' and 'chi2_contingency' from 'scipy.stats', creates a cross-tabulation of gender and product, and then performs a Chi-Square test. The output shows a Chi-Square Value of 11.070894590068523, a P-Value of 0.04999233067024619, and an expected table. The notebook interface includes a file explorer on the left, a terminal at the bottom, and a status bar at the very bottom.

```
import pandas as pd
from scipy.stats import chi2_contingency

table = pd.crosstab(df['gender'], df['product'])
chi2, p, dof, expected = chi2_contingency(table)

print("Chi-Square Value:", chi2)
print("P-Value:", p)
print("Expected Table:\n", expected)
```

[12] ✓ 0.0s Python

... Chi-Square Value: 11.070894590068523
P-Value: 0.04999233067024619
Expected Table:
[[150.17085427 117.76884422 92.84422111 115.89949749 117.14572864
26.170854271

Insight: If $p < 0.05$, product preference depends on gender.



Final Assignment

- 1 Perform t-test for rating between Male and Female.
- 2 Perform Chi-Square test for payment method and gender.